Lab Final

Instructions, background, & Rubric

1. Objectives: Design, construct, and fly your own paper airplane to determine two key aspects of its flight dynamics:

- The upward lift force acting on its wings.
- The energy lost during flight, determined from changes in the airplane's kinetic and potential energy.

2. Equipment List:

Quantity	Item
1	Piece of paper
1	Camera
1	Tracker software and computer

3. Background:

Paper airplanes are simple models of flight that experience a combination of forces as they traverse through the air. The primary forces acting on a paper airplane include:

- Gravity (mg), which pulls the airplane downward
- Aerodynamic drag, which opposes the motion through the air
- Aerodynamic lift, the upward force generated mainly by the wing's shape and its angle of attack relative to the airflow.

The lift force arises due to differences in air pressure above and below the wings, an effect often explained by Bernoulli's principle and the deflection of air. In steady flight, the net vertical forces must balance so that the airplane can either sustain flight or follow a predictable curved trajectory. By recording the flight with a camera and analyzing the vertical motion using Tracker, you can determine the vertical acceleration. Since the airplane's net vertical acceleration a_{net} results from the difference between the upward lift force F_{lift} and the downward gravitational force F_g, we can express this as:

$$ma_{net} = F_{lift} - F_g \quad \rightarrow \quad ma_{net} = F_{lift} - mg \quad \rightarrow \quad F_{lift} = m(g + a_{net})$$

In addition to the force analysis, energy considerations allow us to understand how non-conservative forces (like drag) do work on the airplane during its flight. The kinetic energy (KE) and gravitational potential energy (PE) of the airplane is given by:

$$KE = \frac{1}{2}m(v_x^2 + v_y^2) \qquad PE = mgy$$

By measuring the initial (at launch) and final (at landing or at a defined point in the flight) values for velocity and height, you can calculate the total mechanical energy at each point. The work done by non-conservative forces (W_{nc}), which represents energy lost from the system, is then determined by the difference between the final and initial mechanical energies

$$W_{nc} = E_f - E_i$$

4. Experimental Procedure:

Using video analysis software (Tracker), you will extract the airplane's x and y position and velocity data throughout its flight. First, by analyzing the vertical motion and applying Newton's second law, you will calculate the net upward force (lift) acting on the airplane using:

$$F_{lift} - F_g = ma_y$$

Then, by measuring the initial and final velocities (using both x and y components) as well as the corresponding vertical positions, you will compute the change in the airplane's total mechanical energy and determine the work done by non-conservative forces (e.g., aerodynamic drag) during the flight.

Your airplane's flight should be as horizontal and consistent as possible. If your plane nosedives or jerks around then it will make analysis very, very challenging.

5. Requirements:

Your execution of experiment and group work will be checked and graded by the instructor/TA/LA in class. Complete one lab report for your group using the following rubric to ensure you receive all the points. **Each person should submit a PDF file of the lab report.**

Group Number:

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https://docs.google.com/document/d/1tfeDSTOYp7SGgrF-HM53yVH-L1vgjTlEUMOESK5nRig/edit?usp=drive_link

6. Rubric

Component	Absent (0–5)	Lacking (6–10)	Good (11–15)	Excellent (16–20)
Data Table of Measured Quantities	Incomplete or missing table; key measurements (positions, velocities, time stamps, etc.) not recorded.	Table is present but contains some missing or unclear measurements; units or significant figures may be inconsistent.	Most relevant measurements are included with correct units; minor clarity or organization issues are present.	Complete, well-organized table with all relevant measurements (initial/final x & y positions, velocities, times, mass) and proper units throughout.
Graph of y-Position vs Time and Fit Equation	Graph is missing or very poorly executed;	Graph is present with axis titles and scale, but the graph is using irrelevant data.	Graph is missing either the trendline or the axis titles. Correct trendline is chosen for data	Clear, professionally presented graph with correctly labeled axes, an accurate trendline, and a clearly shown fit equation on the graph.
Analysis of Fit Equation: Acceleration and Lift Force Calculation	Little or no analysis provided; acceleration and force calculations are missing or largely incorrect.	Basic analysis is performed; some attempt to extract acceleration from the fit equation exists, but with significant errors or missing steps.	Analysis is mostly complete; acceleration is correctly determined and the lift force calculated using $F_{lift} - F_g = ma_y$, though some explanations or units may be unclear.	Thorough and accurate analysis with clear, step-by-step reasoning; correct calculations and proper units are provided throughout.
Energy Analysis: Position and Velocity Data	Energy analysis is absent or incorrect; insufficient use of the collected data.	Basic energy analysis is attempted; partial calculations of initial and final kinetic and potential energies are made, but work done by non-conservative forces is not clearly determined.	Complete energy analysis with correct calculation of kinetic ($\frac{1}{2}m(v_x^2+v_y^2)$) and potential (mgy) energies; explanations may lack clarity in places.	Detailed and accurate energy analysis with clear calculations of initial and final mechanical energies, leading to an accurate determination of the work done by non-conservative forces (energy loss).
Final Write-Up: Summary, Error Analysis, Recommendations	Write-up is missing or extremely brief with little to no discussion of results, errors, or improvements.	Write-up summarizes results but lacks depth; minimal discussion of error sources and vague improvement suggestions are provided.	Clear write-up that includes a summary of results, discussion of potential error sources, and some recommendations for improvement; some explanations may be underdeveloped.	Well-written (2–3 clear paragraphs) summary with a thorough discussion of findings, critical error analysis, and insightful, specific recommendations for future improvements.

Lab Report

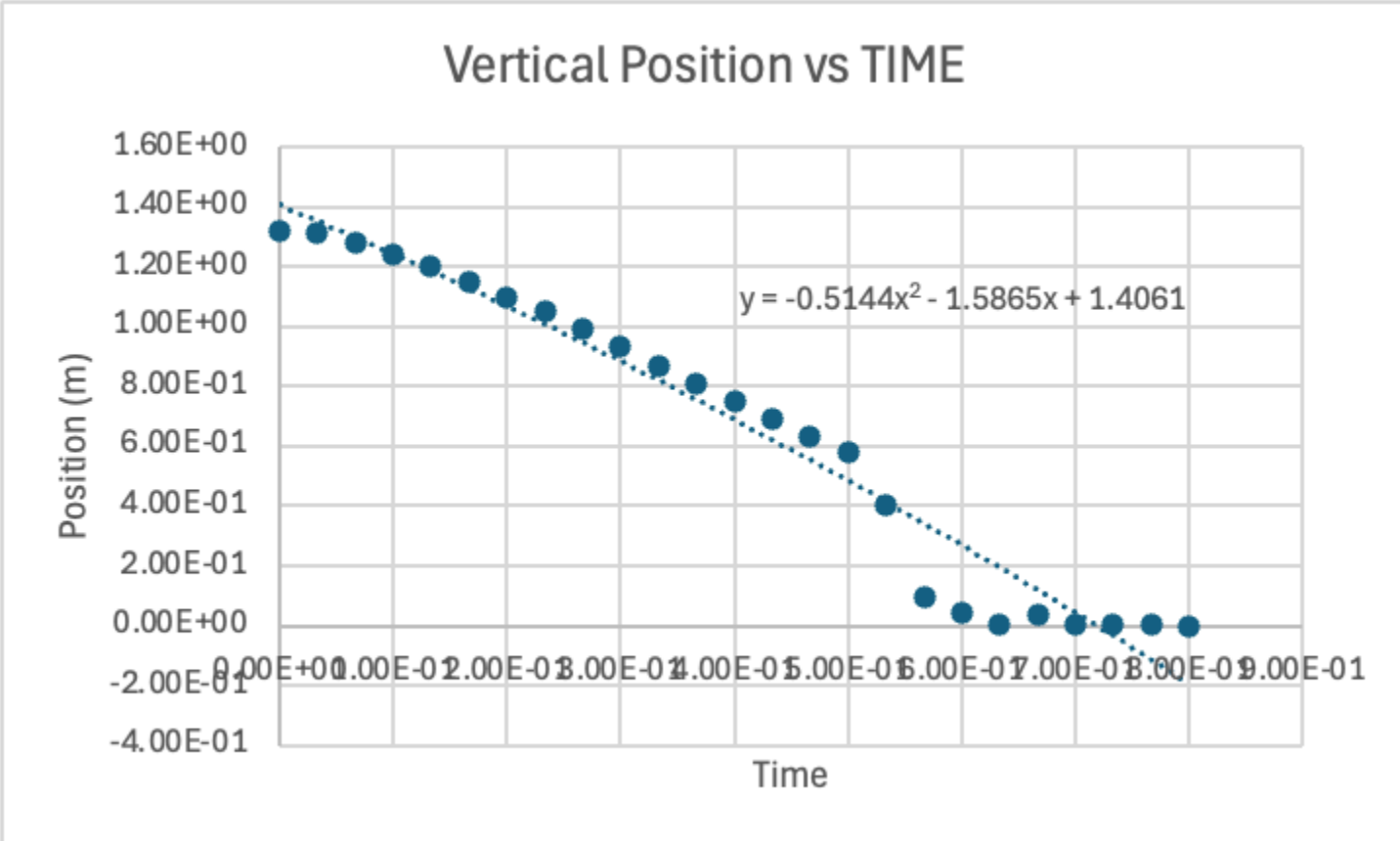
1. **Data Table of Measured Quantities**

time	x	y	vx	vy	ax
0.00E+00	0.00E+00	1.20E+00	0.00E+00	0.00E+00	0.00E+00
3.33E-02	5.26E-02	1.19E+00	3.65E+00	-9.87E-02	0.00E+00
6.67E-02	1.97E-01	1.19E+00	4.14E+00	-2.96E-01	-4.23E+00
1.00E-01	3.29E-01	1.17E+00	3.55E+00	-4.93E-01	-5.92E+00
1.33E-01	4.34E-01	1.16E+00	3.45E+00	-5.92E-01	-3.38E+00
1.67E-01	5.59E-01	1.13E+00	3.85E+00	-7.89E-01	0.00E+00
2.00E-01	6.91E-01	1.11E+00	3.75E+00	-5.92E-01	-1.69E+00
2.33E-01	8.09E-01	1.09E+00	3.36E+00	-7.89E-01	-6.77E+00
2.67E-01	9.14E-01	1.05E+00	3.45E+00	-1.18E+00	-8.46E+00
3.00E-01	1.04E+00	1.01E+00	3.65E+00	-1.28E+00	8.46E-01
3.33E-01	1.16E+00	9.67E-01	3.26E+00	-1.18E+00	3.38E+00
3.67E-01	1.26E+00	9.34E-01	2.86E+00	-9.87E-01	0.00E+00
4.00E-01	1.35E+00	9.01E-01	3.16E+00	-1.18E+00	-5.08E+00
4.33E-01	1.47E+00	8.55E-01	3.75E+00	-1.38E+00	-3.38E+00
4.67E-01	1.60E+00	8.09E-01	3.26E+00	-1.38E+00	8.46E+00
5.00E-01	1.68E+00	7.63E-01	2.47E+00	-8.88E-01	4.23E+00
5.33E-01	1.76E+00	7.50E-01	2.47E+00	-8.88E-01	8.46E-01
5.67E-01	1.85E+00	7.04E-01	2.86E+00	-1.09E+00	8.46E-01
6.00E-01	1.95E+00	6.78E-01	3.16E+00	-6.91E-01	4.23E+00
6.33E-01	2.06E+00	6.58E-01	2.76E+00	-7.89E-01	-5.08E+00
6.67E-01	2.14E+00	6.25E-01	2.07E+00	-1.09E+00	-4.23E+00

7.00E-01	2.20E+00	5.86E-01	2.07E+00	-1.09E+00	8.46E-01
7.33E-01	2.28E+00	5.53E-01	2.37E+00	-9.87E-01	8.46E+00
7.67E-01	2.36E+00	5.20E-01	1.88E+00	-5.92E-01	0.00E+00
8.00E-01	2.40E+00	5.13E-01	2.07E+00	-7.89E-01	-3.38E+00
8.33E-01	2.49E+00	4.67E-01	2.27E+00	-1.09E+00	-2.54E+00
8.67E-01	2.55E+00	4.41E-01	2.17E+00	-7.89E-01	8.46E+00
9.00E-01	2.64E+00	4.14E-01	2.47E+00	-5.92E-01	0.00E+00
9.33E-01	2.72E+00	4.01E-01	2.27E+00	-6.91E-01	-7.61E+00
9.67E-01	2.79E+00	3.68E-01	2.47E+00	-1.18E+00	0.00E+00
1.00E+00	2.88E+00	3.22E-01	2.47E+00	-7.89E-01	6.77E+00
1.03E+00	2.95E+00	3.16E-01	2.37E+00	-4.93E-01	-2.54E+00
1.07E+00	3.04E+00	2.89E-01	2.47E+00	-1.09E+00	-7.61E+00
1.10E+00	3.12E+00	2.43E-01	2.47E+00	-1.09E+00	-8.46E-01
1.13E+00	3.20E+00	2.17E-01	2.17E+00	-9.87E-01	6.77E+00
1.17E+00	3.26E+00	1.78E-01	1.97E+00	-7.89E-01	0.00E+00
1.20E+00	3.34E+00	1.64E-01	2.17E+00	-7.89E-01	-1.69E+00
1.23E+00	3.41E+00	1.25E-01	2.66E+00	-1.09E+00	2.54E+00
1.27E+00	3.51E+00	9.21E-02	2.07E+00	-5.92E-01	1.35E+01
1.30E+00	3.55E+00	8.55E-02	1.09E+00	-9.87E-02	0.00E+00

Initial (x)	Final (x)	Initial (y)	Final (y)	Initial velocity(x)	Final velocity(x)	Initial velocity(y)	Final velocity(y)	Acceleration(y)	Mass of paper airplane (kg)
0	3.546	1.197	8.552	0 m/s	1.085 m/s	0 m/s	-9.868 m/s	0 m/s^2	0.0045 kg

2. **Graph of y-Position vs Time and Fit Equation**



3. **Analysis of Fit Equation: Acceleration and Lift Force Calculation**

Fit equation: **$y = -0.5144x^2 - 1.5865x + 1.4061$**

Net vertical force on plane:

m * a(vertical) = F(lift) - F(g)

0.0045kg* (-0.514 *2) = F(lift) - (0.0045kg * 9.8m/s^2)

-0.00463 N + (0.0441) = F(lift)

F(lift) = **0.0394 N**

By calculating the vertical acceleration from the polynomial trendline of vertical position vs time…we found the net force on the paper airplane. We calculated lift by doing the Net force MINUS the downward force of gravity. The calculated force of lift was 0.0394N.

4. **Energy Analysis: Position and Velocity Data**

Initial energy of the system: $\frac{1}{2}(m * v_i^2) + mgh_{initial} = \frac{1}{2}(0.0045 * (0)^2) + (0.0045(9.8)(1.197))$ = **0.0527877 J**

Final Energy of the system: $\frac{1}{2}(m * v_f^2) + mgh_{final} = \frac{1}{2}(0.0045 * (1.085)^2) + (0.0045(9.8)(0)) =$ **0.00264875625J**

Energy lost: -0.05 J

5. Final Write-Up: Summary, Error Analysis, Recommendations

This experiment measured the upward lift force on the plane, and the total change in the system's mechanical energy from non-conservative forces, such as air resistance or friction. Essentially, the experiment measured how non-conservative forces, specifically air resistance, impacted the object's mechanical energy on its path. The tracker was used to record the plane's kinematics and to calculate the total mechanical energy throughout the flight.

We found that the lift force on the plane was about 0.039N. The overall net force on the object was NEGATIVE, as the force of gravity was stronger than the lift force. The acceleration of the plane was about -1.03 m/s^2(hence why it eventually fell to the floor).

We also notice that the total mechanical energy of the system DECREASED, due to air resistance, drag, etc(the non-conservative forces). The total loss in the system's energy from the beginning to the end of the flight was -0.05 J.

Some errors that may have arisen in this experiment could be the faulty design of the airplane. There may have also been interference from the air vents, which could have caused a flow of air not accounted for.

If we were to mess up this experiment and had to redo it again, we would have to make sure we are using a stronger focus on consistency and accuracy for our plane. Making sure that the airplane is flying as smoothly and horizontally as possible, since sharp turns or a nose-dive can mess up our tracker. We also need to make sure that we are using the right graphs when using the data we have collected from the tracker. Using the wrong table graph can give us incorrect information when trying to find the velocities and accelerations from it.